QING WU PH.D.

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Research interests

Medical Foundation Model, Medical Reconstruction, Medical Multimodality

Work Experience

2026 – PRESENT **Ant Group, Medical AI Lab** Zhejiang, China
– Position: Senior Algorithm Engineer (Ant Star, Plan A Talent Program)
– Advisor: Dr. Le Lu

Education Experience

2020 – 2026 **ShanghaiTech University** Shanghai, China
– Ph.D. in *Electronic Science and Technology*
– Advisor: Prof. Yuyao Zhang

2016 – 2020 **China University of Geosciences, Wuhan** Hubei, China
– B.S. in *Communication Engineering*

Selected Publications

MedIA Haoyuan Le*, Qing Wu*, Bowen Li, Yuyao Zhang, Hongjiang Wei, "Unsupervised high-resolution 3D MRI motion correction via physics-informed implicit neural representations", *Medical Image Analysis*

IEEE TRPMS Qing Wu, Xu Guo, Lixuan Chen, Yanyan Liu, Wuwei Ren, Yifeng Zhang, S Kevin Zhou, Jingyi Yu, Yuyao Zhang, "Solving Energy-Independent Density for CT Metal Artifact Reduction via Neural Representation", *IEEE Transactions on Radiation and Plasma Medical Sciences*

AAAI 2026 Qing Wu, Hongjiang Wei, Jingyi Yu, Yuyao Zhang, "Unsupervised Multi-Parameter Inverse Solving for Reducing Ring Artifacts in 3D X-Ray CBCT", *40th Annual AAAI Conference on Artificial Intelligence*

ICLR 2025 Qing Wu*, Chenhe Du*, XuanYu Tian, Jingyi Yu, Yuyao Zhang, Hongjiang Wei, "Moner: Motion Correction in Undersampled Radial MRI with Unsupervised Neural Representation", *13th International Conference on Learning Representations,* ***Spotlight Paper, Top 5%***

NEURIPS 2023 Qing Wu, Lixuan Chen, Ce Wang, Hongjiang Wei, S. Kevin Zhou, Jingyi Yu, Yuyao Zhang, "Unsupervised Polychromatic Neural Representation for CT Metal Artifact Reduction", *37th Conference on Neural Information Processing Systems*

IEEE TCI — Qing Wu, Ruimin Feng, Hongjiang Wei, Jingyi Yu, Yuyao Zhang, "Self-Supervised Coordinate Projection Network for Sparse-View Computed Tomography", *IEEE Transactions on Computational Imaging,* ***Top 25 Most Downloaded Article, Sept. 2023 – Sept. 2024***

IEEE ISBI 2023 — Qing Wu, Xin Li, Hongjiang Wei, Jingyi Yu, Yuyao Zhang, "Joint Rigid Motion Correction and Sparse-View CT via Self-Calibrating Neural Field", *IEEE 20th International Symposium on Biomedical Imaging*

IEEE TCAS-I — Chaolin Rao*, Qing Wu*, Pingqiang Zhou, Jingyi Yu, Yuyao Zhang, Xin Lou, "An Energy-Efficient Accelerator for Medical Image Reconstruction From Implicit Neural Representation", *IEEE Transactions on Circuits and Systems I: Regular Papers*

IEEE J-BHI — Qing Wu, Yuwei Li, Yawen Sun, Yan Zhou, Hongjiang Wei, Jingyi Yu, Yuyao Zhang, "An Arbitrary Scale Super-Resolution Approach for 3D MR Images via Implicit Neural Representation", *IEEE Journal of Biomedical and Health Informatics*

MICCAI 2021 — Qing Wu, Yuwei Li, Lan Xu, Ruimin Feng, Hongjiang Wei, Qing Yang, Boliang Yu, Xiaozhao Liu, Jingyi Yu, Yuyao Zhang, "IREM: High-Resolution Magnetic Resonance Image Reconstruction via Implicit Neural Representation", *24th International Conference on Medical Image Computing and Computer Assisted Intervention,* ***Early Accepted, Top 13%***

Teaching Experience

FALL 2022/2023 — **CS276: Computational Photography** — Shanghaitech University
– Position: Guest Lecture
– Instructor: Prof. Jingyi Yu

SPRING 2023 — **CS270B: Advanced Digital Image Processing** — Shanghaitech University
– Position: Teaching Assistant
– Instructor: Prof. Yuyao Zhang

Academic Service

REVIEWER — **Conferences:** *ICLR, NeurIPS, ICML, ECCV, CVPR, AAAI, MICCAI, IEEE ISBI*
Journals: *IEEE TIP, MedIA, IEEE TMI, IEEE TNNLS, IEEE TCI, IEEE J-BHI, JCDE*